

Brandon Davis
Senior Data Engineer

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Professional Summary:

- ✓ Senior Data Engineer with 10+ years of IT experience, Designed and implementing Cloud ETL solutions using **Azure Synapse Analytics, Azure Databricks, and Azure Data Factory** and Data Warehousing Solutions.
- ✓ **6 years** of dedicated experience in **Azure Cloud Services** and **Big Data technologies**, designing and establishing optimal cloud solutions for efficient data migration and processing.
- ✓ Hands - on experience in **Azure Cloud Services, Azure Synapse Analytics, SQL Azure, Data Factory, Azure Analysis services, Application Insights, Azure Monitoring, Key Vault, Azure Data Lake.**
- ✓ Created **Azure Data Factory** (ADF) Batch pipelines to Ingest data from relational sources into Azure Data Lake Storage (**ADLS Gen2**) & incremental fashion and then load into Delta tables after cleansing.
- ✓ Experience with Azure transformation projects and implement ETL and data movement solutions using **Azure Data Factory.**
- ✓ Involved in development of roadmaps and deliverables to advance the migration of existing solutions on-premises systems/applications to **Azure Cloud.**
- ✓ Designed and built a configurable, file-based and stream-based data ingestion framework to load data from multiple sources into bronze and silver layers, cutting down coding time by 40%
- ✓ Developed Data pipelines, **Dataflows, Event Streams, Lakehouse, ML Models, Notebooks, Reports, Spark jobs** and warehouse in Synapse Data engineering on **Microsoft Fabric.**
- ✓ Experience in Developing Spark applications using **Spark - SQL** in **Databricks** for data extraction, transformation and aggregation from multiple file formats for analyzing & transforming the data to uncover insights into the customer usage patterns.
- ✓ Developed real-time streaming solutions using **Apache Kafka** and **Azure Event Hubs**, integrated with **Azure Databricks** for stream processing using **PySpark.**
- ✓ Implemented ad-hoc analysis solutions using **Azure Data Lake Analytics/Store, HDInsight.**
- ✓ Implemented data replication and synchronization across **Snowflake** and various workflow levels, employing Change Data Capture techniques to enhance data integrity. Achieved a 98% reduction in data inconsistencies.
- ✓ Strong understanding of **Data Modelling** and **ETL process** in data warehouse environment such as **star schema, snowflake schema.**
- ✓ Extensive knowledge and experience in dealing with **Relational Database Management** Systems Normalization, Stored Procedures, Constraints, Joins, Indexes, Data Import/Export, Triggers.
- ✓ Expert at data transformations like lookups, Derived Column, Conditional Splits, Sort, Data Conversion, Multicast and Derived columns, Union All, Merge Joins, Merge, Fuzzy Lookup, Fuzzy Grouping, Pivot, Un - pivot and SCD to load data in SQL SERVER destination.
- ✓ Extensive experience in developing applications that perform Data Processing tasks using Teradata, Oracle, SQL Server, and MySQL databases.
- ✓ Experience in Extraction, Transformation and Loading (**ETL**) data from various sources into Data Warehouses, as well as data processing like collecting, aggregating, and moving data from various sources using **Apache Flume, Kafka.**
- ✓ Developed iterative algorithms using **Spark Streaming** in Scala and Python to builds near real-time dashboards.
- ✓ Experienced in loading data to Hive partitions and created buckets in **Hive** and developed **Map Reduce** jobs to automate transfer the data from **H Base.**
- ✓ Worked with various file formats such as **CSV, JSON, XML, ORC, Avro, and Parquet file formats.**
- ✓ Worked on data processing and transformations and actions in spark by using Python or PySpark language.
- ✓ Acquired experience in Spark scripts in **Python, Scala, and SQL** for advancement in development and examination through analysis.

- ✓ Strong working knowledge across the technology stack including **ETL, data analysis**, data cleansing, data matching, data quality, audit, and design.
- ✓ Extensive experience in designing and tuning **OLAP and OLTP** systems, ensuring efficient processing of transactional and analytical workloads for high-performance database environments.
- ✓ Created interactive **dashboards and reports in Power BI**, leveraging data from various sources for insightful business intelligence and reporting.
- ✓ Scheduled and monitored data workflows with **Control-M and Apache Airflow** for coordinated execution of complex tasks.
- ✓ Involved in all the phases of **Software Development Life Cycle** (Requirements Analysis, Design, Development, Testing, Deployment, and Support) and **Agile methodologies**.

Technical Skills:

Azure Services	Azure Data Factory, Azure Data Bricks, Logic Apps, Functional App, Azure DevOps, Synapse Analytics, Key Vault, Azure Event Hub, Azure Monitor, Azure Blob Storage, Azure Functions, MS Fabric.
Databases	Snowflake, MS SQL Server 2016/2014/2012, ADLS Gen2, Azure SQL DB, MS Excel, MS Access, Oracle 11g/12c, Cosmos DB, PostgreSQL, MySQL
Big Data Technologies	MapReduce, Hive, Teg, Python, PySpark, Scala, Kafka, Spark streaming, Oozie, Sqoop, HDFS, Flume, Zookeeper
Hadoop Distribution	Cloudera, Horton Works, Apache
Languages	Python, SQL, PL/SQL, Java, HiveQL, Scala, Shell Scripting, Bash, PowerShell
Web Technologies	HTML, CSS, JavaScript, XML, JSP, Restful, SOAP
Operating Systems	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS
Build Automation tools	Ant, Maven, Jenkins
Version Control	GIT, GitHub, Bitbucket
Methodology	Agile, Scrum
IDE & Build Tools, Design	Eclipse, Visual Studio, PyCharm, IntelliJ IDEA, VS Code

Education:

- Masters in Computer Science at University of Texas at Austin in 2013
- Bachelors in Computer science at University of Texas at Dallas in 2011

Professional Experience:

Client: Hone Health

July 2024 to Present

Role: Azure Data Engineer

Responsibilities:

- ✓ Designed and implemented end-to-end ETL pipelines using **Azure Data Factory**, incorporating **Copy Activities, Mapping Data Flows**, and **Control Flow Activities**, improved in data transferring efficiency.
- ✓ Optimized data transformation workflows with **Azure Databricks**, leveraging PySpark DataFrames, RDDs, and Databricks Runtime to enhance data processing performance by 60%.
- ✓ Developed complex SQL scripts in **Azure Synapse Analytics**, implementing columnstore indexes and PolyBase for external data source integration to enhance query performance.
- ✓ Utilized **Massively Parallel Processing (MPP)** architecture within Azure Synapse to handle large datasets and optimize critical reporting dashboards.
- ✓ Implemented Delta Lake architecture on **ADLS Gen2**, leveraging Hierarchical Namespace and RBAC for secure and efficient data management.
- ✓ Implemented **Delta Live Tables** for continuous data processing and utilized **Databricks SQL** for interactive querying.

- ✓ Implemented **ADLS Gen2** Lifecycle Management policies and Immutable Storage for **Data governance**, resulting in a scalable and cost-effective analytics platform.
- ✓ Integrated Functional Apps within **Azure Data Factory** pipelines for automated data processing tasks, enhancing overall system flexibility and responsiveness.
- ✓ Developed real-time data ingestion solutions using Azure Data Factory integrated with **Azure Event Hubs**, improving data synchronization accuracy and reducing latency.
- ✓ Implemented data integration and transformations within **Azure SQL Server**, ensuring high availability and performance for enterprise-level applications.
- ✓ Designed and implemented scalable data governance frameworks using **Databricks Unity Catalog** to manage permissions, audit data usage, and ensure compliance with industry standards across multiple environments, improving data security and operational efficiency by 30%.
- ✓ Worked with **Cosmos DB** to deliver cloud-native data services optimized for performance, reliability, and scalability.
- ✓ Created workspaces in **Fabric** and sharing with team as common development environment.
- ✓ Designed and developed end to end solutions for data ingestion, data processing and visualization with help of Fabric capabilities.
- ✓ hands-on experience in unified development on **Fabric (SaaS)** with Fabric by using Data Factory, Data Science, Data Warehousing, Data Engineering, Data activator, Power BI, Real Time Analytics and One Lake.
- ✓ Developed and optimized complex **T-SQL** queries for data manipulation and reporting in Azure SQL Server, enhancing data retrieval efficiency.
- ✓ Implemented **Azure Monitor** for comprehensive monitoring of data pipelines, enabling proactive identification and resolution of performance bottlenecks.
- ✓ Implemented **Azure Key Vault** to securely manage secrets, keys, and certificates within Azure Data Factory pipelines, significantly strengthening security and ensuring compliance with industry standards.
- ✓ Configured **Linked Services** in Azure Data Factory to seamlessly connect and integrate various data sources, improving data flow and integration efficiency.
- ✓ Created and managed **Datasets** within Azure Data Factory, ensuring accurate data mapping and transformation across multiple data sources and destinations.
- ✓ Experienced in configuring **Linked Services**, creating **Datasets**, and setting up **Integration Runtime** within Azure Data Factory for seamless data integration across diverse environments.
- ✓ Implemented advanced scheduling and automation using Triggers like **Scheduled Trigger**, **Tumbling Window Trigger** and **Event Based Trigger** based on necessity in Azure Data Factory, ensuring timely execution of ETL processes and data synchronization.
- ✓ Implemented **Tumbling Window Triggers** and **Linked Services** to connect multiple data sources, reducing data latency and processing time for large-scale data integrations.
- ✓ Designed and developed data models in **Snowflake**, leveraging its MPP architecture for scalable and high-performance data storage and querying.
- ✓ Implemented Snowflake's **Time Travel** and **Zero-Copy Cloning** features for efficient data backup and recovery, reducing downtime and data loss risks.
- ✓ Developed and maintained real-time data streaming pipelines using **Apache Kafka**, enabling efficient data flow and processing for real-time analytics.
- ✓ Integrated Kafka with **Azure Event Hubs** for seamless data streaming between cloud and on-premise systems, improving data synchronization and reducing latency.
- ✓ Optimized data pipelines and **Spark jobs** in **Azure Databricks** through advanced techniques like Spark configuration tuning, data caching, and data partitioning, resulting in superior performance and efficiency.
- ✓ Developed automated data quality checks using **PySpark** in **Azure Databricks**, significantly reducing data errors and enhancing overall data trustworthiness.
- ✓ Leveraged **PySpark** for large-scale data processing tasks within Databricks, improving data processing performance and reducing execution times.
- ✓ Developed and optimized **SQL queries** for data extraction, transformation, and loading (ETL) processes across various platforms, ensuring data accuracy and efficiency.

- ✓ Managed and optimized data storage and processing using various file formats Like **PARQUET, JSON, CSV**, ensuring compatibility and efficiency across multiple systems.
- ✓ Utilized specific file formats such as **Parquet** and **ORC** for efficient storage and querying in big data environments, improving data retrieval speeds.
- ✓ Developed and optimized complex **Snow SQL queries** for efficient data extraction, transformation, and loading (ETL) operations within **Snowflake**.
- ✓ Designed and implemented **Snowflake Schema** for optimized data warehousing, enabling efficient data storage and retrieval.
- ✓ Developed **Star Schema** data models for OLAP systems, ensuring efficient query performance and simplified data navigation.
- ✓ Developed interactive **Power BI dashboards** to visualize key business metrics, improving decision-making processes and enabling real-time insights for stakeholders.
- ✓ Integrated Power BI with Azure Synapse Analytics for real-time data visualization, reducing the time to insight by 40%.
- ✓ Worked in a fast-paced **Agile environment**, collaborating with cross-functional teams to deliver high-quality data solutions **on time and within budget**.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen2, Azure Blob, Snowflake Data Warehouse, Azure Event Hubs, Azure Functions, Azure Logic Apps, Function Apps, Azure Monitor, Cosmos DB, Azure Synapse Analytics, Azure DevOps, Azure Fabric, Azure Key Vault, MS SQL, Oracle, Python, PySpark, JIRA, Kafka, Power Bi.

Client: DXC Technology, Chicago, IL

Jan 2021 to June 2024

Role: Azure Data Engineer

Responsibilities:

- ✓ Designed and implemented complex data integration pipelines in **Azure Data Factory**, utilizing various activities including Copy Activity, Mapping Data Flow, and Control Flow Activities.
- ✓ Configured **Linked Services** to connect to multiple data sources and implemented Triggers for automated execution.
- ✓ Implemented **Azure Data Factory's** Integration Runtime to enable secure data movement between on-premises and cloud environments. Implemented **Self-Hosted Integration Runtime** for accessing data behind corporate firewalls.
- ✓ Optimized Data storage and management in **Azure Data Lake Storage Gen2**, implementing Hierarchical Namespace for efficient data organization.
- ✓ Leveraged **Azure Databricks** Workspace to create and manage clusters, developing notebooks for data processing and analysis.
- ✓ Developed ETL processes using **Azure Databricks**, leveraging Spark DataFrames and RDDs for efficient data transformation.
- ✓ Implemented **Azure Data Lake Storage Gen2** with **Azure Active Directory** (AAD) integration, ensuring seamless and secure authentication and authorization for robust data access control.
- ✓ Created robust data pipelines in **Azure Synapse Analytics**, using Data Flows for code-free ETL and Notebooks for complex transformations.
- ✓ Developed data transformation and analysis scripts in **Python**, leveraging libraries like **Pandas** and **NumPy** for efficient data manipulation and processing.
- ✓ Implemented end-to-end analytics solutions using **Azure Synapse Analytics** Workspace, combining SQL Pools for data warehousing and Spark Pools for big data processing.
- ✓ Leveraged **Azure Synapse** Link to enable near real-time analytics on operational data, integrating with **Azure Cosmos DB** for seamless data synchronization.
- ✓ Implemented One time Data Migration of Multistate level data from SQL server to **Snowflake** by using Python and **SnowSQL**.
- ✓ Designed and implemented data ingestion pipelines using **Sqoop** and **Flume** to efficiently import structured and unstructured data from various on-premises systems into **HDFS** and **Azure Blob Storage**.
- ✓ Configured Kafka to stream real-time data from multiple sources into **Hadoop** and Azure environments, ensuring low-latency data processing.
- ✓ Developed complex data transformation workflows using **Apache Pig, Hive, and Spark** to process and cleanse large datasets, ensuring data quality and consistency.

- ✓ Employed **MapReduce and YARN** to optimize distributed data processing, significantly reducing job execution times for large-scale data transformations.
- ✓ Migrated and managed large-scale datasets in **HDFS, Azure Data Lake, and Apache Cassandra**, optimizing storage strategies for performance and cost-efficiency.
- ✓ Leveraged **Cosmos DB and HBase for NoSQL** data storage, enabling high-performance read and write operations for various applications.
- ✓ Automated end-to-end data workflows using **Oozie and Control-M**, ensuring seamless data movement and processing across hybrid cloud environments.
- ✓ Analyzed **Zookeeper** for distributed coordination and consistency management across **Hadoop and Azure ecosystems**.
- ✓ Conducted extensive performance tuning of Spark jobs, **SQL queries**, and **Hadoop clusters**, achieving significant improvements in data processing speed and resource utilization.
- ✓ Implemented best practices for data partitioning, indexing, and caching to enhance the performance of **Hive queries** and **SQL Server databases**.
- ✓ Secured data pipelines by integrating **Azure Key Vault** for managing secrets, keys, and certificates, ensuring compliance with organizational security policies.
- ✓ Managed and optimized data storage and processing using various file formats Like **ORC, PARQUET, AVRO, JSON, CSV**, ensuring compatibility and efficiency across multiple systems.
- ✓ Collaborated with data analysts to design and develop scalable reporting solutions using **Hive and Spark**, providing actionable insights from processed data.
- ✓ Enabled seamless **Data Visualization and reporting** through integration with **Power BI**, driving data-driven decision-making across the organization.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen 2, Azure Blob, Logic Apps, Functional App, Snowflake, MS SQL, Oracle, HDFS, Hive, Sqoop, PIG, PySpark, Oozie, Zookeeper, SQL, Python, Scala, Kafka, Power BI.

Client: Intuitive, Sunnyvale, CA

Jan 2018 to Nov 2020

Role: Big Data Engineer

Responsibilities:

- ✓ Developed and optimized complex **MapReduce** programs for processing large datasets, improving performance by 30%.
- ✓ Built and maintained **HDFS structures**, ensuring efficient storage and retrieval of petabyte-scale data.
- ✓ Created **Hive queries** and optimized **HiveQL** scripts for data extraction, transformation, and loading (ETL), reducing query execution time by 25%.
- ✓ Managed and automated data pipelines using **Apache Oozie**, orchestrating multi-step workflows across the **Hadoop ecosystem**.
- ✓ Integrated **Apache Kafka** with **Hadoop** for real-time data streaming and processing, enhancing data ingestion speeds by 40%.
- ✓ Designed and implemented **Sqoop** jobs for seamless data transfer between Hadoop and relational databases like **MS SQL Server and MySQL**.
- ✓ Developed **PySpark** and Scala scripts to process large-scale data, achieving a 20% increase in processing efficiency.
- ✓ Maximized **Hadoop** and **Spark cluster** performance by fine-tuning configurations, optimizing resource allocation, and reducing job execution times by up to 40%.
- ✓ Optimized **Apache Cassandra** for distributed database management, achieving high availability and seamless scalability of data storage across multiple nodes.
- ✓ Deployed **Apache Zookeeper** for distributed coordination, improving reliability and fault tolerance across **Hadoop** clusters.
- ✓ Engineered data ingestion workflows using **Apache Flume**, enabling efficient and reliable log data streaming to **HDFS**.
- ✓ Performed data modeling and query optimization in HBase, leading to faster data retrieval in low-latency applications.
- ✓ Implemented **Control-M** for scheduling and monitoring batch jobs, ensuring timely execution of critical data processing tasks.
- ✓ Developed **Pig** scripts for ETL processes, simplifying complex data transformations and improving data pipeline robustness.

- ✓ Designed and implemented **Cosmos DB** solutions for scalable, globally distributed data storage, enhancing application performance.
- ✓ Conducted performance tuning on **Hive and Spark** jobs, achieving significant improvements in processing speed and resource utilization.
- ✓ Automated cluster management tasks using **YARN**, reducing operational overhead and improving cluster efficiency.
- ✓ Leveraged **Python and YAML** for configuration management and scripting, streamlining deployment processes and reducing errors.
- ✓ Integrated **Apache Spark** with HDFS for distributed data processing, optimizing computational efficiency for **Big Data analytics**.
- ✓ Collaborated with cross-functional teams to implement end-to-end data solutions, leveraging **Hadoop** and related technologies to meet business requirements.

Environment: Sqoop, Pig, HDFS, Apache Cassandra, Oozie, Zookeeper, Control-M, Flume, Kafka, Apache Spark, Scala, Hive, Hadoop, HBase, Cosmos DB, MySQL, YAML, GIT, JIRA

Client: *Mercury Insurance, Austin, TX*

Oct 2014 to Dec 2017

Role: *Data Warehouse Developer*

Responsibilities:

- ✓ Designed and managed SQL Server databases, ensuring efficient data storage, retrieval, and overall database performance.
- ✓ Led data modeling efforts, performing both physical and logical database design to support complex business requirements.
- ✓ Integrated front-end applications with SQL Server backend, enhancing data flow and application functionality.
- ✓ Developed complex T-SQL scripts, including stored procedures, triggers, indexes, and user-defined functions, to automate tasks and optimize database performance.
- ✓ Executed data migration projects using DTS, facilitating the seamless transfer of data between servers and different environments.
- ✓ Optimized T-SQL queries for enhanced performance, reducing query execution times and improving system responsiveness.
- ✓ Leveraged SSIS/DTS to transfer and transform data from diverse sources like MS Excel, MS Access, and flat files into SQL Server, utilizing advanced features like data conversion and derived columns.
- ✓ Created and deployed sophisticated SSIS packages for ETL processes, with logging mechanisms at both package and task levels to track data processing activities.
- ✓ Provided 24/7 production support, developing and testing SQL Server queries and Windows command files to monitor and ensure database health and availability.
- ✓ Designed and generated complex reports using SQL Reporting Services, including cross-tab, drill-down, and OLAP reports, tailored to meet specific business needs.

Environment: SQL Server, Erwin, TOAD, Windows 2000, Oracle 9i, UNIX Shell Scripts, PL/SQL, Data Transformation Services (DTS), SQL Server Integration Services (SSIS), MS Excel, MS Access, SQL Server Reporting Services (SSRS).